Factor Premia since 1963, and Do Factor ETFs Deliver Them?

Investment Research Portfolio · Project 7 · Factor data July 1963 – August 2026

Summary

Over 758 months from July 1963 to August 2026, the US market and momentum premia were the largest (7.2% and 7.2% a year, t = 3.7 and 3.9). Value, profitability and investment earned about 3% a year each and are significant over the full sample, while size (2.2%, t = 1.7) is not. Every factor had long losing stretches: value lost 56% from January 2007 to December 2020 and has since recovered 43% of that loss. An equal-weight blend of value, profitability, investment and momentum had a Sharpe ratio of 0.74, against 0.49 for the best single factor. In Part B, all five iShares factor ETFs load significantly on their target factor, but the size of the exposure differs widely: MTUM carries 0.45 units of the momentum factor and VLUE 0.42 units of value, while QUAL (0.17 on profitability) and SIZE (0.14 on size) are close to the market portfolio with a small tilt. No ETF has a significant alpha.

1. Data

- **Factors:** Fama/French 5 Factors (2x3) and the Momentum Factor, monthly, from the Kenneth French Data Library, July 1963 to August 2026 (758 months; August 2026 is the latest month published at the time of writing). Mkt-RF is the value-weighted US market minus the one-month T-bill; SMB (small minus big), HML (high minus low book-to-market), RMW (robust minus weak profitability), CMA (conservative minus aggressive investment) and Mom (winners minus losers over months t–12 to t–2) are long-short portfolios.
- **Macro:** NBER recession indicator (FRED USREC) and CPI for all urban consumers (FRED CPIAUCSL). FRED has no CPI value for October 2025 because the Bureau of Labor Statistics did not publish one during the government shutdown, so that month is excluded from the inflation split only.
- **ETFs:** iShares MSCI USA Momentum Factor (MTUM), Quality Factor (QUAL), Value Factor (VLUE), Min Vol Factor (USMV) and Size Factor (SIZE). Daily adjusted closes from yfinance converted to monthly total returns. Regressions run from August 2013 to August 2026 (157 months), ending with the factor data.

Part A. Sixty-three years of factor returns

A1. Premium, risk and significance

Factor	Premium /yr	Volatility	Sharpe	t-stat	Max drawdown	Worst 5y /yr
Mkt-RF	7.2%	15.4%	0.47	3.72	-55.8%	-11.5%
SMB	2.2%	10.5%	0.21	1.70	-56.4%	-10.9%
HML	3.5%	10.3%	0.34	2.72	-57.8%	-10.7%
RMW	3.0%	7.9%	0.38	3.03	-41.8%	-5.8%
CMA	2.9%	7.2%	0.41	3.23	-27.6%	-3.8%
Mom	7.2%	14.6%	0.49	3.90	-57.8%	-11.8%

Table 1. Factor statistics, Jul 1963 – Aug 2026. Premium = mean monthly return × 12; volatility × √12; t-stat = mean ÷ (standard deviation ÷ √T). Drawdowns and the worst rolling 5-year return use compounded returns.

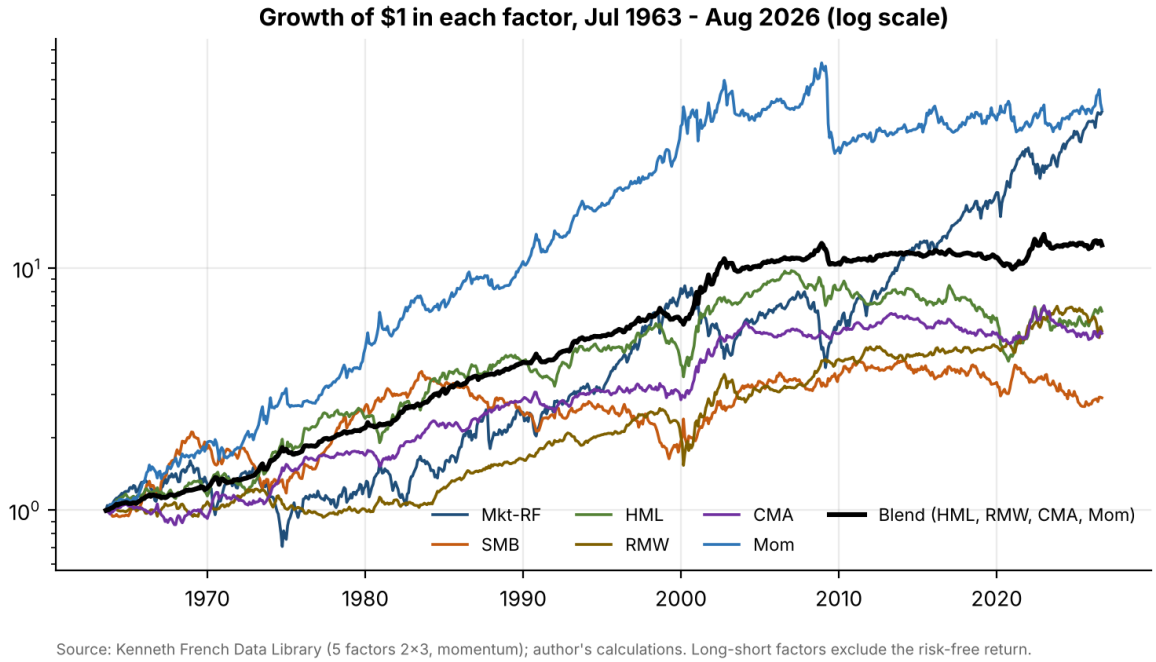
The market and momentum premia are the largest and most significant, at about 7% a year each. Momentum's t-statistic of 3.9 is the highest of the six. Value, profitability and investment each earned around 3% a year with t-statistics between 2.7 and 3.2. Size is the weakest: 2.2% a year with a t-statistic of 1.70, below the conventional 2.0 threshold, so over this sample it is not distinguishable from zero. Factor investing is not a smooth ride: four of the six factors suffered drawdowns above 55%, and all six had at least one five-year stretch with a negative compounded return.

A2. Decade by decade

Decade	Mkt-RF	SMB	HML	RMW	CMA	Mom
1960s (from Jul 1963)	3.9%	9.5%	2.1%	1.3%	-1.2%	10.5%
1970s	-0.3%	4.1%	7.7%	-0.7%	6.2%	9.3%
1980s	7.3%	-0.7%	5.6%	4.9%	5.5%	8.6%
1990s	12.5%	-2.5%	-0.4%	2.2%	-0.1%	13.7%
2000s	-3.1%	6.5%	7.4%	7.6%	6.5%	-2.0%
2010s	13.0%	-0.7%	-2.6%	1.2%	0.1%	2.6%
2020s (to Aug 2026)	12.2%	-2.2%	1.0%	2.2%	-0.1%	1.5%

Table 2. Annualized compounded return of each factor by decade.

Premia come in long waves. Value earned 5–8% a year in the 1970s, 1980s and 2000s but lost money in the 1990s and 2010s. Momentum was strong in every decade until the 2000s, when the 2009 momentum crash pushed it negative. The 2010s and 2020s have been unusually poor for every long-short factor while the market itself earned over 12% a year, which explains much of the frustration with factor products in that period.



Source: Kenneth French Data Library (5 factors 2×3, momentum); author's calculations. Long-short factors exclude the risk-free return.

Figure 1. Growth of $1 in each factor and in the equal-weight blend, log scale. Long-short factors are shown without the T-bill return.

A3. Correlations

Factor	Full Mkt-RF	Full SMB	Full HML	Full RMW	Full CMA	Full Mom	2010+ Mkt-RF	2010+ SMB	2010+ HML	2010+ RMW	2010+ CMA	2010+ Mom
Mkt-RF	1.0	0.27	-0.21	-0.19	-0.36	-0.16	1.0	0.33	0.03	-0.14	-0.16	-0.25
SMB	0.27	1.0	0.02	-0.33	-0.08	-0.08	0.33	1.0	0.33	-0.31	0.08	-0.28
HML	-0.21	0.02	1.0	0.1	0.68	-0.19	0.03	0.33	1.0	0.1	0.63	-0.26
RMW	-0.19	-0.33	0.1	1.0	0.02	0.06	-0.14	-0.31	0.1	1.0	0.19	-0.11
CMA	-0.36	-0.08	0.68	0.02	1.0	-0.02	-0.16	0.08	0.63	0.19	1.0	-0.0
Mom	-0.16	-0.08	-0.19	0.06	-0.02	1.0	-0.25	-0.28	-0.26	-0.11	-0.0	1.0

Table 3. Correlation matrix of monthly factor returns, full sample (left) and January 2010 – August 2026 (right).

HML and CMA are the most closely related pair (0.68 full sample, 0.63 since 2010): value firms tend to be conservative investors, which is why Fama and French (2015) find HML largely redundant once CMA and RMW are included. Momentum is negatively correlated with value (-0.19; -0.26 since 2010), which is the

basis of the classic value-momentum combination. Since 2010, SMB has become more correlated with HML (0.33 vs. 0.02) and the market.

A4. Regimes: recessions and high inflation

Factor	Recession	Expansion	t (diff.)	CPI > 4%	CPI ≤ 4%	t (diff.)
Mkt-RF	-0.61%	0.76%	-1.81	-0.03%	0.92%	-2.60
SMB	0.17%	0.19%	-0.05	0.16%	0.20%	-0.21
HML	0.29%	0.29%	0.00	0.48%	0.21%	1.20
RMW	0.35%	0.24%	0.43	0.28%	0.25%	0.17
CMA	0.73%	0.18%	1.91	0.44%	0.15%	1.71
Mom	0.04%	0.67%	-0.85	0.83%	0.48%	1.14

Table 4. Average monthly factor returns by regime. NBER recessions: 85 of 758 months. Year-on-year CPI inflation above 4%: 255 of 757 months. t = Welch t-statistic for the difference.

The market premium is much lower in recessions (-0.61% vs. 0.76% a month) and when inflation is above 4% (-0.03% vs. 0.92%; t = -2.60). None of the long-short factors has a premium that differs significantly between regimes: the largest effects are CMA doing better in recessions (t = 1.91) and momentum doing worse (t = -0.85). With only 85 recession months, these splits have little statistical power, so the safest reading is that the style factors are not reliable recession hedges, but they also did not lose money on average when the market did.

A5. Value's lost decade

From January 2007 to December 2020, HML compounded at -5.7% a year, a cumulative loss of 55.8%, with the trough in September 2020. For comparison, HML earned 5.4% a year from July 1963 to December 2006. Explanations offered in the literature include the rising importance of intangible assets that book value does not capture, falling interest rates that favoured long-duration growth stocks, and crowding. From January 2021 to August 2026, HML returned 53.8% cumulatively. Because a 56% loss needs a 126% gain to recover, this has made back 43% of the dollar loss: value has recovered strongly, but not fully.

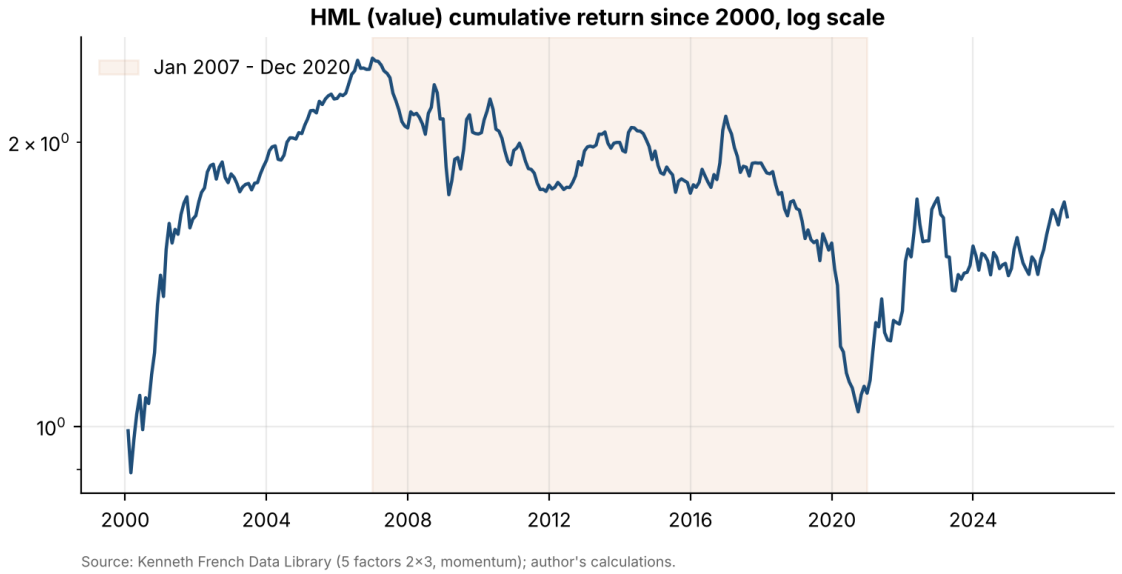


Figure 2. Cumulative HML return since 2000, log scale; shaded: January 2007 – December 2020.

A6. A multi-factor blend

Portfolio	Premium /yr	Volatility	Sharpe	Max drawdown
HML	3.5%	10.3%	0.34	-57.8%
RMW	3.0%	7.9%	0.38	-41.8%
CMA	2.9%	7.2%	0.41	-27.6%
Mom	7.2%	14.6%	0.49	-57.8%
Blend	4.2%	5.6%	0.74	-21.8%

Table 5. Equal-weight blend of HML, RMW, CMA and Mom (rebalanced monthly) vs. each factor alone.

Because the four factors are weakly or negatively correlated, combining them cuts volatility to 5.6% while keeping a 4.2% premium. The Sharpe ratio rises to 0.74, about 1.5 times the best single factor, and the maximum drawdown falls to -21.8%. Diversifying across factors is the most reliable way to harvest them.

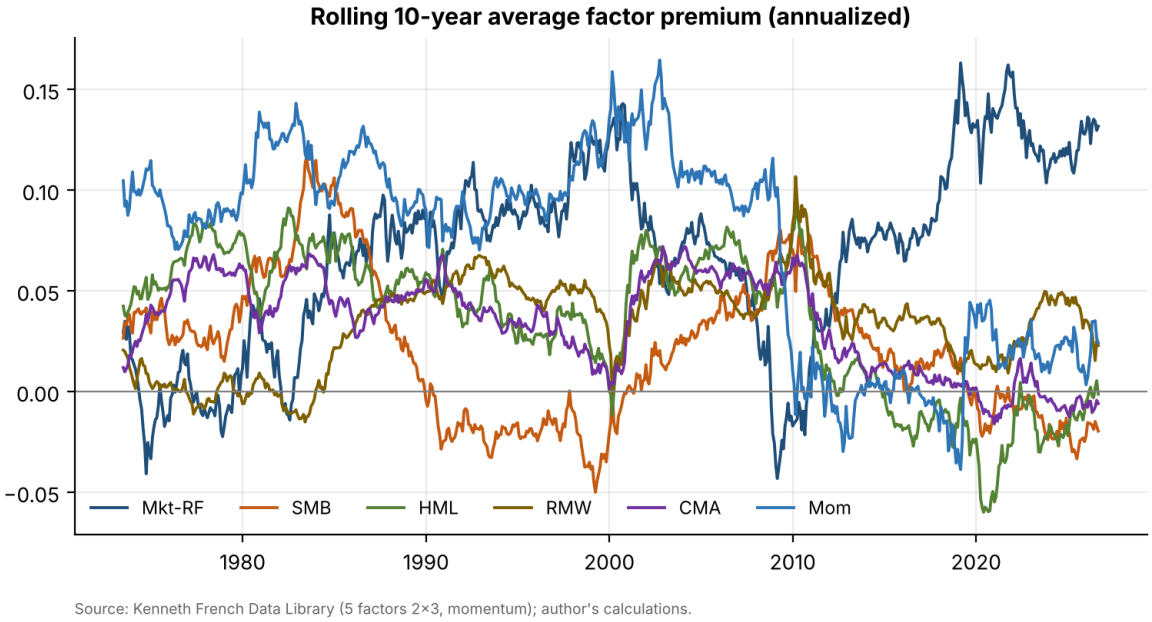


Figure 3. Rolling 10-year average premium of each factor. Every factor, including the market, has had decade-long windows with a negative average premium.

Part B. Do factor ETFs deliver their factor?

For each ETF I regress the monthly return minus the one-month T-bill on the six factors, August 2013 – August 2026: $r - rf = \alpha + \beta_M(\text{Mkt–RF}) + s\cdot\text{SMB} + h\cdot\text{HML} + p\cdot\text{RMW} + c\cdot\text{CMA} + m\cdot\text{Mom} + \varepsilon$. Standard errors are Newey-West with three lags. A loading of 1 on a long-short factor would mean the ETF carries as much of that factor as the Fama-French research portfolio itself.

ETF	Alpha /yr	t(α)	Mkt-RF	SMB	HML	RMW	CMA	Mom	R^2
MTUM	-0.5%	-0.35	1.10 (29.2)	-0.12 (-2.3)	-0.01 (-0.2)	-0.26 (-2.3)	0.08 (0.8)	0.45 (8.5)	0.89
VLUE	0.2%	0.11	1.05 (15.7)	-0.04 (-0.4)	0.42 (7.3)	-0.29 (-1.5)	0.08 (1.1)	-0.03 (-0.5)	0.87
QUAL	-0.6%	-0.80	0.98 (49.0)	-0.07 (-2.3)	-0.03 (-0.9)	0.17 (4.2)	0.03 (0.5)	0.02 (0.9)	0.96
SIZE	-1.1%	-1.15	0.95 (41.2)	0.14 (3.9)	0.12 (2.8)	0.07 (1.6)	0.04 (0.6)	-0.03 (-1.1)	0.94
USMV	-0.6%	-0.36	0.72 (16.5)	-0.07 (-1.3)	-0.03 (-0.4)	0.27 (4.3)	0.20 (2.0)	0.07 (1.9)	0.79

Table 6. Six-factor regressions; loadings with Newey-West t-statistics in brackets.

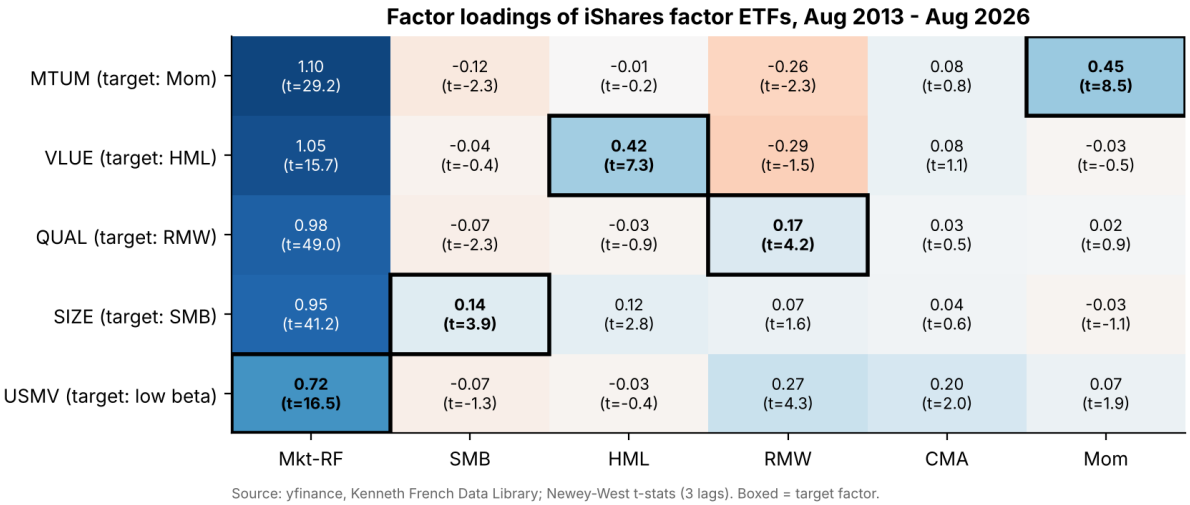


Figure 4. Headline result: factor loadings of the five ETFs. The boxed cell is the factor each ETF is named after. For USMV the target is a market beta below 1 (t vs. 1 = -6.4).

ETF	Target	Loading	t-stat	Largest style loading	Delivers?
MTUM	Mom	0.45	8.46	Mom	True
VLUE	HML	0.42	7.35	HML	True
QUAL	RMW	0.17	4.16	RMW	True
SIZE	SMB	0.14	3.88	SMB	True
USMV	Mkt-RF < 1	0.72	-6.40	RMW	True

Table 7. Is the target factor the ETF's largest exposure? For MTUM, VLUE, QUAL and SIZE the comparison is among the five style factors (every long-only equity fund has a market beta near 1). For USMV the t-stat tests $\beta = 1$.

All five ETFs deliver a statistically significant exposure to their target, and for MTUM, VLUE, QUAL and SIZE it is their largest style exposure. The magnitudes differ a lot. MTUM (0.45) and VLUE (0.42) carry close to half of a pure long-short factor, which is a lot for a long-only fund. QUAL (0.17) and SIZE (0.14) are mostly market exposure: their R^2 values are 0.96 and 0.94 and market betas are near 1. USMV delivers what it promises, a market beta of 0.72, and it also has a larger profitability loading (0.27) than QUAL itself: low-volatility stocks tend to be profitable, conservative firms. None of the five ETFs has an alpha significantly different from zero after accounting for these exposures.

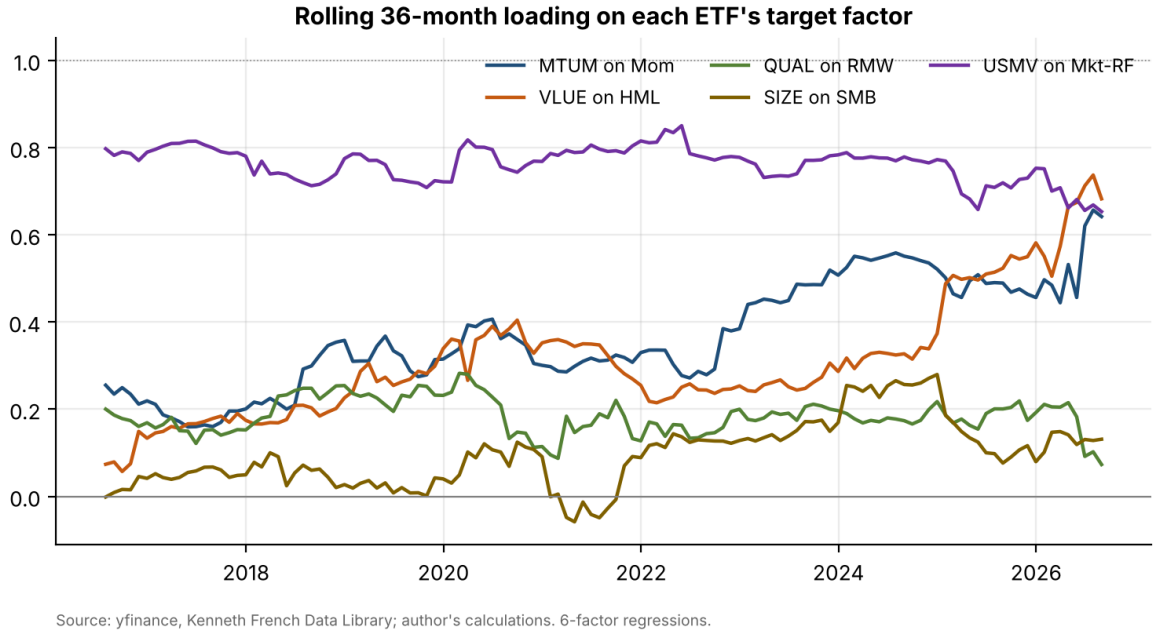


Figure 5. Rolling 36-month loading on each ETF's target factor.

Exposures change over time. VLUE's value loading rose from about 0.07 in the first 36-month window to 0.68 in the latest. MTUM's momentum loading rose from about 0.26 to 0.64. QUAL's profitability loading has stayed small and was 0.07 in the latest window, and SIZE's size loading was briefly negative in 2021. An investor buying these funds for a factor needs to monitor the exposure, not only the label.

Fund	Ann. return	Volatility	Sharpe
MTUM	15.3%	16.7%	0.83
VLUE	13.2%	17.8%	0.69
QUAL	13.7%	14.6%	0.84
SIZE	11.6%	15.3%	0.68
USMV	10.8%	11.5%	0.80
Market (Mkt-RF + RF)	14.0%	14.9%	0.84

Table 8. ETF performance, Aug 2013 – Aug 2026, against the Fama-French market portfolio.

Over the same period no ETF had a higher Sharpe ratio than the market portfolio (0.839); QUAL (0.836) and MTUM (0.832) came closest. This is consistent with the weak long-short factor returns of the 2010s and 2020s in Table 2.

ETF – model	Alpha /yr	t(α)	Market beta	R^2
MTUM – CAPM	2.0%	0.90	0.95	0.73
MTUM – FF3	1.2%	0.59	0.98	0.75
VLUE – CAPM	-0.8%	-0.31	1.04	0.77
VLUE – FF3	-0.3%	-0.21	1.04	0.85
QUAL – CAPM	0.3%	0.32	0.95	0.94
QUAL – FF3	-0.3%	-0.35	0.98	0.95
SIZE – CAPM	-1.8%	-1.33	0.98	0.91
SIZE – FF3	-1.2%	-1.20	0.96	0.94
USMV – CAPM	1.0%	0.58	0.66	0.72
USMV – FF3	0.2%	0.10	0.70	0.74

Table 9. Robustness: CAPM and Fama-French three-factor regressions (Newey-West t-statistics).

Under the CAPM and the three-factor model the conclusions hold: no ETF has a significant alpha (largest |t| = 1.33), and USMV's beta stays well below 1 (0.66 under the CAPM).

Conclusion

Sixty-three years of data support a market, momentum, value, profitability and investment premium, but not a reliable size premium, and every factor has gone through losing periods of five years or more. The practical answer to these drawdowns is diversification across factors, which raised the Sharpe ratio of the best single factor by about half. The iShares factor ETFs do deliver the factor on the label, but in very different doses: momentum and value funds carry meaningful exposure, while the quality and size funds are close to the market with a modest tilt, so their names overstate what investors get.

Limitations

- Fama-French factors are paper long-short portfolios before trading costs; investors cannot capture them in full.
- The ETF sample is only 157 months and dominated by one long bull market.
- Recession and inflation splits have few observations and low statistical power.
- Factor definitions in the ETFs' MSCI indices differ from Fama-French definitions; loadings measure overlap, not intent.

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